

Students Tim Mullins and Brian Figueiredo and Assistant Professor Eric Meyer monitor data generated by the movements of Bill Slesinski in Lawrence Tech's Experimental Biomechanics Laboratory.



The Science of **MOVEMENT**

Introducing Lawrence Tech's new Experimental Biomechanics Laboratory

The College of Engineering's new Experimental Biomechanics Laboratory, developed by one of Lawrence Tech's newest faculty members (see page XX), will advance experimental biomechanics understanding by providing practical training to students and expanding the knowledge of biomechanics through applied research.

"One of the lab's focus areas is on real world measuring of human movement, a skill that can be used in many applications – from understanding the gait cycle of a child with cerebral palsy to optimizing the golf swing of a player in the PGA, or simulating how a customer would interact with a new automotive interior design," explained lab director Eric

Meyer, assistant professor of biomedical engineering.

Meyer can set up sophisticated motion capture equipment in a large open space in the Engineering Building. Six Vicon Bonita cameras, manufactured by industry leader Vicon Industries, are mounted on tripods and positioned around the "capture volume" area at strategic locations. The cameras are connected to an interface box that can support up to 10 cameras and other data sources, such as force plates and EMG (muscle activation recording) systems.

After setup, a calibration is performed to "teach" the cameras their relative positions in the room; then they can be used to measure the three-dimensional position

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of reflective markers attached to a person or object with an accuracy rate within 1 millimeter. The cameras are also able to record at 240 frames per second, which is almost 10 times faster than standard video cameras.

Students in the Biomechanics Laboratory course will find the lab a valuable resource for applying the theories and concepts they learn in their Introduction to Biomechanics lectures to the practice of measuring and analyzing human movement. This course, required for juniors in the biomedical engineering program, will be introduced in the coming fall semester.

Equipment in the Experimental Biomechanics Laboratory will have a number of other uses as well, such as biomechanical research into the joint injuries experi-

enced by athletes. This subject was the basis for Meyer's doctoral research at Michigan State University.

Meyer is currently investigating ACL injury in the knee during a jump landing as well as high ankle sprains caused by rotation that can occur when a football player's cleats get caught in the playing surface – the former with a Lawrence Tech faculty seed grant and the latter with a grant from the NFL charities.

He is also working with the game art program in Lawrence Tech's College

of Architecture and Design, teaching a studio module on computer-generated animation using motion capture – the same technique used to create the movie "Avatar" as well of hundreds of popular video games.

"Biomechanics is one of the oldest fields of research, but there are also very interesting and cutting-edge topics," he said. "I love how engineering tools can be applied to solve medical problems and to understand ourselves."

Powerful electron microscope will enhance teaching and research

Thanks to a \$769,444 grant from the National Science Foundation, the College of Engineering is now the home to the Detroit area's first environmental scanning electron microscope.

The powerful microscope can be used to scientifically characterize wet, oily, porous, and soft materials that are traditionally considered impossible to study using conventional microscopy techniques, according to Assistant Professor Yawen Li, the principal investigator for the NSF grant.

This will greatly enhance a variety of research activities conducted at Lawrence Tech and its collaborating institutions in biomaterials, orthopedics, tissue engineering, construction materials, automotive components, and lithium ion batteries.

In addition to its use as both a teaching and research tool at Lawrence Tech, the new microscope, which was delivered in February, also will be used for research projects at Wayne State University, Oakland University, the William Beaumont Research Institute, and the General Mo-

tors Research and Development Center.

The microscope's credentials are impressive, if not downright mind-boggling. It can produce high-resolution images of a sample surface, revealing details less than a nanometer (one-billionth of a meter) in size. To put this into perspective, a very fine human hair is about 10,000 nanometers wide. The maximum magnification of more than 500,000 times is about 250 times the magnification limit of the best light microscopes.

"The environmental scanning electron



Students Stacy Kaiser and Erick Blank receive instruction from Assistant Professor Yawen Li on how to operate Lawrence Tech's new environmental scanning electron microscope.

microscope will strengthen the existing materials research infrastructure at Lawrence Tech and our partnering institutions, and enhance the research and educational experiences of a large number of students at Lawrence Tech and other universities," Li said.

Philip Olivier discusses a problem with students Haley Smith and Ryan Gilmore in a control lab.

His research interests include steel connection design; fatigue and fracture behavior of structural steel connections and components; erection engineering; effects of heat treatments on structural properties, serviceability, and load capacities of steel beam bridges; and the design, behavior, and analysis of structures subjected to fire and seismic and blast loading. His current research work focuses on the material properties of composite materials and fire, smoke, and toxicity properties of materials.

Drawing on this background, Kowalkowski is the author of numerous articles and technical reports for professional publications and government agencies.

Kowalkowski received his PhD in structural engineering from Purdue University and a Bachelor of Science in Civil Engineering from Michigan State University. He has taught courses on steel design, concrete design, the theory of structures, and advanced structural analysis.

Kun Hua

Assistant Professor, Electrical and Computer Engineering



Kun Hua

Hua joins Lawrence Tech from the Department of Computer and Electronics Engineering at the University of Nebraska-Lincoln, where he was a postdoctoral research associate and instructor. His research interests include wireless communications, wireless sensor networks, digital signal processing, and embedded system design.

Hua's recent research has focused on software defined radios (SDR) and digital processing. Using SDR techniques, he is trying to minimize the amount of hard-

ware used in a radio system by moving analog-to-digital and digital-to-analog converters as close to the radio frequency front end as possible.

This research will benefit both users and service providers by offering nearly unlimited flexibility with reduced hardware costs and power usage. Hua also plans to extend his research to developing an adaptive vehicle-embedded system for radio content detection.

Hua received his doctorate in computer and electronic engineering from the University of Nebraska-Lincoln and his bachelor's and master's in electrical and computer engineering from the University of Xi'an Jiaotong in China.

Liping Liu

Assistant Professor, Mechanical Engineering



Liping Liu

Liu's current research interests are thermal sciences and energy systems, with a special interest in addressing transport phenomena in energy processes.

Most of her work is directed toward improving performance and efficiency of various types of energy generation, conversion, and consumption applications.

Her research background includes wind-tunnel experiments, and theoretical analysis of the impact of surface wettability on heat transfer performance, and energy efficiency of heat exchangers. She also has studied the performance improvement and energy efficiency of heat exchangers using vortex generation techniques.

At Lawrence Tech, Liu plans to extend her specialization in thermal sciences and engineering to the development and improvement of high-performance, environmentally friendly energy systems. She hopes to contribute in both enhancing current applications and advancing the implementation of non-conventional/emerging technologies.

Liu earned her doctorate in mechanical engineering from the University of Illinois at Urbana-Champaign and her bachelor's in mechanical engineering at the University of Science and Technology of China.

Eric Meyer

**Assistant Professor, Biomedical Engineering
Director, Experimental Biomechanics Laboratory**



Eric Meyer

Meyer's goal in establishing the new Experimental Biomechanics Laboratory at Lawrence Tech is to advance biomechanics understanding by providing practical training to students and advancing knowledge through translational research.

To achieve these goals, he combines in vivo testing of human and animal subjects with cadaver and surrogate in vitro models to understand injury mechanisms. Ultimately, by collaborating with clinicians, he aims to develop preventative and regenerative treatments for soft tissue damage and degeneration.

Passionate about how engineering tools can be applied to solve medical problems, Meyer was recently a guest speaker at a session on "Solving the Mystery of Non-contact ACL Injuries – How Do They Happen?" at the International Olympic Committee World Conference on Prevention of Injury & Illness in Sport, held in Monte Carlo.

Meyer earned his doctorate in mechanical engineering from Michigan State University, where his research focused on lower extremity biomechanics during sports injuries. He then completed a postdoctoral fellowship in cartilage tissue at the Centre for Bioengineering at Trinity College Dublin in Ireland.